

BREAST CANCER ULTRASOUND CLASSIFICATION USING MACHINE LEARNING

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An auditable study of preprocessing, convolutional models, and evidence
integrity

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By

Akerele David Damilola

Department of Computing and Mathematics

Evidence status: provisional

The supplied cohort fails the subject-level split audit; no
clinical-performance claim is made.

Abstract

Breast ultrasound is a useful adjunct to mammography, but its interpretation is operator dependent and image appearance is affected by speckle, acquisition settings, and lesion context. This dissertation develops a reproducible research prototype for binary classification of breast-ultrasound images using convolutional neural networks. The software integrates contrast-limited adaptive histogram equalisation (CLAHE), aspect-ratio-preserving square padding, transfer-learning backbones (EfficientNet-B0 and ResNet-50), and a custom convolutional baseline. It also provides a FastAPI research demonstration for controlled image inference and visual inspection.

The principal contribution is methodological rather than clinical: a single preprocessing implementation is shared by training, evaluation, command-line inference, and the web service; paired lesion masks are recognised under the naming conventions used by the supplied datasets; and evaluation emits machine-readable metrics, figures, and an audit trail. A read-only data audit records split sizes and flags cross-split subject identifiers before performance is interpreted.

The local OASBUD and BrEaST copies were rebuilt into subject-level partitions and pass the split audit. BUSI remains excluded because its renamed image copy does not retain enough provenance to verify patient separation. A fresh EfficientNet-B0 checkpoint evaluated on 62 held-out BrEaST and OASBUD images produced 67.74% accuracy, macro F1 0.6774, and ROC-AUC 0.7419. These values describe a local research cohort, not clinical or external validation. Historical headline figures, pseudo-labelling ablations, and noise curves contained in legacy project assets are therefore not treated as empirical findings.

The dissertation records the corrected workflow: rebuild the cohort at subject level, preserve source identifiers and preprocessing manifests, retrain with fixed seeds, and generate every reported table and figure from held-out predictions. The system remains an educational research prototype and is not a diagnostic or clinical decision-support device.

Declaration

No part of this project has been submitted in support of an application for any other degree or qualification at this or any other institute of learning. Apart from those parts of the project containing citations to the work of others, this project is my own unaided work. This work has been carried out in accordance with the Manchester Metropolitan University research ethics procedures, and has received ethical approval number 92016.

Signed: Akerele David Damilola

A handwritten signature in black ink, appearing to read 'Akerele', written over a light gray rectangular background.

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Abbreviations

ABUS	Automated Breast Ultrasound System
ACR	American College of Radiology
AI	Artificial Intelligence
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
BI-RADS	Breast Imaging-Reporting and Data System
BUS	Breast Ultrasound
BUSI	Breast Ultrasound Images Dataset
CAD	Computer-Aided Diagnosis
CBAM	Convolutional Block Attention Module
CDF	Cumulative Distribution Function
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNR	Contrast-to-Noise Ratio
CNN	Convolutional Neural Network
ECE	Expected Calibration Error
FDA	United States Food and Drug Administration
FPR	False Positive Rate
GLCM	Gray-Level Co-occurrence Matrix
Grad-CAM	Gradient-weighted Class Activation Mapping
HOG	Histogram of Oriented Gradients
MBConv	Mobile Inverted Bottleneck Convolution
MDR	Medical Device Regulation (European Union)
OASBUD	Open Access Ultrasound Database
PDF	Probability Density Function
POCUS	Point-of-Care Ultrasound
PR-AUC	Precision-Recall Area Under Curve
RF	Radiofrequency
ROI	Region of Interest
SaMD	Software as a Medical Device
SE	Squeeze-and-Excitation
SNR	Signal-to-Noise Ratio
SRAD	Speckle Reducing Anisotropic Diffusion
SVM	Support Vector Machine
TGC	Time Gain Control
TPR	True Positive Rate (Sensitivity)
ViT	Vision Transformer

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Chapter 1

Introduction

1.1 Background

Breast cancer is a major global health burden and was the most commonly diagnosed cancer worldwide in the 2020 GLOBOCAN estimates [Sung et al. 2021]. Imaging is central to detection and assessment. Ultrasound is particularly useful as a non-ionising adjunct that can characterise breast tissue and guide procedures, but its appearance varies with acquisition settings, equipment, anatomy, and operator technique [Stavros et al. 1995; Al-Dhabyani et al. 2020]. These sources of variation complicate both human interpretation and the development of machine-learning systems.

Convolutional neural networks (CNNs) can learn image representations directly from pixels, and transfer learning can make these models practical when labelled medical datasets are comparatively small [Litjens et al. 2017; He et al. 2016; Tan and Le 2019]. However, strong performance on a curated image split does not by itself establish patient-independent generalisation. Views from the same lesion or subject can be highly correlated, so placing related images in different partitions can inflate an evaluation estimate. Reproducibility therefore depends on data provenance and group-aware partitioning as much as on model architecture.

This dissertation develops a research prototype for binary classification of breast-ultrasound images. The project combines configurable contrast enhancement, aspect-ratio-preserving preprocessing, three CNN backbones, machine-readable evaluation artifacts, a split-integrity audit, and a local web interface. It explicitly separates what the present software demonstrates from what would require a new, audited experiment.

1.2 Problem statement

Three connected problems motivate the work.

1. **Image variability.** Speckle, gain, field of view, and resolution vary between im-

ages and datasets. Preprocessing must be specified and shared across training, evaluation, and inference rather than reimplemented independently.

2. **Geometric transformation.** Directly resizing a rectangular image to a square uses different horizontal and vertical scale factors. This can alter apparent lesion geometry. Square padding before resizing applies one isotropic scale factor, although any effect on classification must be measured rather than assumed.
3. **Evaluation validity.** Image-level stratification is insufficient when subjects contribute multiple views. An evaluation is interpretable only when subject identifiers, partitions, preprocessing, seeds, checkpoints, and per-image predictions are retained.

The supplied local dataset copy exposes the third problem directly: identifiable OAS-BUD and BrEaST subjects cross partitions, while renamed BUSI files do not retain enough provenance to verify patient separation. The current checkpoint can therefore be used to test the software path, but not to estimate clinical or patient-independent performance.

1.3 Aim and research questions

The aim is to design and critically evaluate a reproducible software framework for breast-ultrasound image classification while maintaining a clear boundary between engineering evidence and clinical claims.

The study is organised around the following questions:

- RQ1** Can a single, testable preprocessing and inference path support training, evaluation, command-line prediction, and a web demonstration?
- RQ2** How should CLAHE and aspect-ratio-preserving square padding be evaluated against simpler preprocessing choices without confounding data partitions, model budgets, or random seeds?
- RQ3** What conclusions are justified by the current dataset provenance and split audit, and what additional controls are required for a patient-independent experiment?

RQ1 is addressed through implementation and regression testing. RQ2 defines a controlled ablation protocol; the present local cohort does not support a comparative conclusion. RQ3 is addressed through the repository audit and the evidence-status analysis in Chapter 4.

1.4 Objectives

The project objectives are to:

1. review ultrasound image formation, speckle, relevant lesion descriptors, CNN transfer learning, and reproducible evaluation;
2. provide deterministic, subject-aware data-splitting and read-only audit utilities;
3. implement CLAHE and three explicitly selectable geometric preprocessing strategies;
4. implement EfficientNet-B0, ResNet-50, and a compact custom CNN using a common binary output contract;
5. record training configuration and data-audit status in saved checkpoints;
6. generate summary metrics, per-image predictions, a confusion matrix, and an ROC curve from one evaluator;
7. expose research-only inference, comparison, and controlled noise demonstrations through a local FastAPI application; and
8. keep the dissertation, notebook, documentation, web interface, Word document, and presentation consistent with the same evidence source.

1.5 Contributions

The principal contribution is not a claim of diagnostic superiority. It is an evidence-aware implementation that makes common sources of divergence visible and testable. Specifically, the repository now provides:

- a shared preprocessing function and paired-mask discovery for supported dataset naming conventions;
- a training guard that refuses an unaudited cohort unless the user explicitly requests a provisional run;
- evaluation outputs that include audit status, preprocessing configuration, per-image probabilities, and an audit checksum;
- regression tests for preprocessing dimensions, mask discovery, data-status reporting, and the research-only API boundary; and
- synchronized written and visual artifacts that withdraw unsupported historical benchmark claims.

1.6 Scope and limitations

The software performs binary image classification using 2D B-mode breast-ultrasound images. The configured classes are **benign** and **malignant**; images stored in a **normal** directory are mapped to the non-malignant class, which is a modelling simplification that must be reconsidered in a definitive study. The work does not establish a diagnosis, assign BI-RADS, recommend management, or validate a medical device. Heatmaps, confidence values, morphology descriptors, and image-quality estimates are exploratory model outputs only.

No pseudo-labelling experiment, preprocessing ablation, multi-seed comparison, external-site evaluation, or prospective reader study is claimed as completed. These are appropriate extensions only after the cohort is rebuilt with original identifiers and a frozen subject-level test partition.

1.7 Dissertation structure

Chapter 2 reviews the physical and computational background. Chapter 3 defines the data audit, preprocessing, models, training controls, and evaluation protocol. Chapter 4 reports the current executable result and its evidence limitations. Chapter 5 discusses validity, reproducibility, and clinical boundaries. Chapter 6 concludes and specifies the work required for a defensible final experiment.

Chapter 2

Background and Literature Review

2.1 Breast ultrasound as an imaging modality

Ultrasound forms images from the transmission and reception of acoustic waves. At a simplified level, the speed of a small-amplitude pressure wave in a medium is related to its bulk modulus K and density ρ by

$$c = \sqrt{\frac{K}{\rho}}. \quad (2.1)$$

At an interface between tissues with acoustic impedances Z_1 and Z_2 , the normal-incidence intensity reflection coefficient can be written as

$$R_I = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2. \quad (2.2)$$

Spatially varying reflection, scattering, attenuation, time-gain compensation, and scanner processing jointly determine the displayed B-mode image. The resulting pixel array is therefore not a direct tissue measurement and can vary substantially between acquisitions.

Breast ultrasound is used alongside clinical history and other imaging. Radiologists may consider features such as shape, margin, orientation, echogenicity, and posterior acoustic behaviour within the ACR BI-RADS ultrasound lexicon [[Mendelson et al. 2013](#); [Stavros et al. 1995](#)]. These descriptors provide clinical context for why geometry and local contrast matter. They do not justify inferring a BI-RADS category from this project’s classifier or from its heuristic image measurements.

2.2 Speckle and image variability

Speckle arises from constructive and destructive interference among echoes from unresolved scatterers. A frequently used simplified observation model combines signal-

dependent and additive components:

$$I(x, y) = f(x, y) \eta_m(x, y) + \eta_a(x, y), \quad (2.3)$$

where f is an underlying reflectivity field, η_m is a multiplicative term, and η_a represents additive acquisition noise. This model is an approximation: displayed clinical images also contain beamforming, compression, filtering, and device-specific processing.

Traditional reduction methods include local-statistics filters and speckle-reducing anisotropic diffusion [Lee 1980; Yu and Acton 2002]. More recent work evaluates learned models under perturbations to determine whether apparently strong classifiers remain stable when image quality changes [Jiang et al. 2023]. Robustness must be tested on stored predictions with a specified perturbation process and seed. A visually plausible synthetic example is not by itself a performance experiment.

2.3 Geometric preprocessing

CNN batches usually require a common spatial size. If an image of height H and width W is directly resized to H' by W' , its axis-specific scale factors are

$$s_y = \frac{H'}{H}, \quad s_x = \frac{W'}{W}. \quad (2.4)$$

When $s_x \neq s_y$, the transformation is anisotropic and changes apparent aspect ratios. An alternative is to crop a defined region, pad it to a square, and then resize it. If the square side is d , both axes use the same final factor $224/d$ in this project.

This geometric property is deterministic, but its effect on classification is empirical. Cropping can discard useful context; reflection padding can introduce repeated boundary structure; and mask-guided training can differ from unmasked deployment. A fair ablation must hold the partition, architecture, optimisation budget, and random seeds constant while changing only the preprocessing strategy.

2.4 Contrast Limited Adaptive Histogram Equalisation

Global histogram equalisation uses an image-wide cumulative distribution and can over-emphasise background variation. Contrast Limited Adaptive Histogram Equalisation (CLAHE) instead applies local mappings and clips histogram mass before redistribution [Zuiderveld 1994]. For a contextual region with histogram $h(k)$, a conceptual clipped histogram is

$$h_c(k) = \min(h(k), \beta) + r(k), \quad (2.5)$$

where β is the clip threshold and $r(k)$ distributes the clipped mass. Interpolation between neighbouring mappings reduces tile-boundary discontinuities.

The implementation in this repository applies OpenCV CLAHE to the luminance channel in LAB colour space, with a clip limit of 2.0 and a 16×16 tile grid. This is a documented configuration choice, not evidence that CLAHE improves discrimination. Its effect requires a controlled ablation and should include both predictive and image-quality analysis.

2.5 Convolutional representation learning

CNNs learn filters through repeated convolution, non-linearity, and aggregation. For an activation tensor $\mathbf{X}^{(l-1)}$, a layer can be represented schematically as

$$\mathbf{X}^{(l)} = \sigma(\mathbf{W}^{(l)} * \mathbf{X}^{(l-1)} + \mathbf{b}^{(l)}). \quad (2.6)$$

Medical-image datasets are often much smaller than general computer-vision corpora, so transfer learning is commonly used to initialise useful low- and mid-level features [Litjens et al. 2017].

2.5.1 ResNet-50

Residual networks learn a residual function around an identity path,

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{\mathbf{W}_i\}) + \mathbf{x}, \quad (2.7)$$

which supports optimisation of deeper architectures [He et al. 2016]. In this project, the original classification layer is replaced by dropout and a two-output linear layer.

2.5.2 EfficientNet-B0

EfficientNet uses coordinated scaling of network depth, width, and input resolution, with the B0 model serving as the baseline architecture [Tan and Le 2019]. Its mobile inverted bottleneck blocks include squeeze-and-excitation operations that reweight channels [Hu et al. 2018]. The project replaces the classifier with dropout and a two-output linear layer.

2.5.3 Custom CNN baseline

The custom model provides a compact, from-scratch baseline with four convolutional stages, batch normalisation, ReLU activations, max pooling, adaptive average pooling, dropout, and a two-output layer. It helps distinguish the contribution of transfer learning

from that of the surrounding pipeline, but a fair comparison requires equivalent split integrity and a predeclared tuning budget.

2.6 Outputs, confidence, and explanation

The two logits are converted to a mutually exclusive probability vector using softmax:

$$p(y = c \mid \mathbf{x}) = \frac{\exp(z_c)}{\sum_{j=1}^K \exp(z_j)}. \quad (2.8)$$

Softmax normalisation does not guarantee calibration. Neural networks can be overconfident, so a displayed score must not be interpreted as diagnostic certainty [Guo et al. 2017]. Calibration should be evaluated on an independent validation design if the model is developed further.

Grad-CAM uses gradients flowing into a convolutional layer to form a coarse localisation map [Selvaraju et al. 2017]. Such maps can support debugging and hypothesis generation, but they neither prove causal reasoning nor validate clinical relevance. This project labels them as research visualisations.

2.7 Dataset provenance and evaluation design

The BUSI dataset paper describes 780 images with normal, benign, and malignant labels and associated masks [Al-Dhabyani et al. 2020]. OASBUD includes multiple views and radio-frequency-derived information for breast lesions [Piotrkowska-Wroblewska et al. 2017]. These characteristics make group-level separation essential. When related views cross partitions, a model may encounter subject-specific appearance during both optimisation and evaluation.

Reproducible evaluation therefore requires more than a random seed. A defensible record includes original subject identifiers, grouping rules, dataset version and licence, checksums, exclusions, split manifest, preprocessing configuration, model configuration, training seed, selected checkpoint, and per-image predictions. Only then can summary metrics be independently regenerated.

2.8 Literature synthesis and research gap

The literature establishes several relevant components: public breast-ultrasound datasets [Al-Dhabyani et al. 2020; Piotrkowska-Wroblewska et al. 2017]; deep representation learning for medical images [Litjens et al. 2017; Cheng et al. 2016]; efficient and residual transfer-learning architectures [He et al. 2016; Tan and Le 2019]; contrast enhancement

[Zuiderveld 1994]; and perturbation-based robustness testing [Jiang et al. 2023]. The gap addressed here is practical integration with an explicit evidence boundary. A classifier, dashboard, and polished report can appear complete while still lacking the provenance needed to justify their headline numbers.

Accordingly, this dissertation treats data auditability as part of the modelling method. It implements the full software path, reports the current executable output as provisional, and defines the group-aware experiment required before comparative or clinical conclusions can be drawn.

Chapter 3

Methodology and Experimental Design

3.1 Study design and evidence levels

The project follows an implementation-and-evaluation design. Its primary unit of analysis is an image, but its required unit of partitioning is the subject. Two evidence levels are distinguished:

Engineering evidence confirms that preprocessing, model loading, inference, metric computation, and artifact generation execute consistently.

Generalisation evidence estimates performance on subjects independent of model development and requires verified subject-level separation, preserved provenance, and a frozen test set.

The current repository supports the first level. The repaired BrEaST and OASBUD cohort passes the split audit, while BUSI is excluded until its original identifiers are restored. Results remain local-cohort research evidence rather than clinical validation.

3.2 Data sources and label mapping

The local project combines organised copies of BUSI, OASBUD, and BrEaST. BUSI is a public B-mode image dataset with normal, benign, and malignant labels and lesion masks [Al-Dhabyani et al. 2020]. OASBUD contains multiple views per lesion [Piotrkowska-Wroblewska et al. 2017]. The BrEaST copy uses case identifiers and includes a small normal subset.

The configured task is binary: `benign` and `malignant`. Files in a `normal` directory are mapped to the non-malignant class by the loader. This makes the executable task

consistent but collapses clinically different categories, so the decision must be reviewed and predeclared before a definitive experiment.

The validated loader finds the counts in Table 3.1. Mask files are excluded from the image count; BUSI is retained for provenance context but excluded from validated training.

Table 3.1: Images in the repaired validated local folders. BUSI is excluded because its patient identifiers are unavailable.

Dataset	Training	Validation	Test	Total
BUSI	–	–	–	–
OASBUD	138	30	30	198
BrEaST	164	33	32	229
Validated combined	302	63	62	427

3.3 Split-integrity audit

The read-only audit enumerates eligible image files, derives subject identifiers where the naming convention permits, and records every partition associated with an identifier. For OASBUD it removes the view suffix; for BrEaST it uses the `caseNNN` prefix. The BUSI files in this local copy were renamed to generated sample identifiers, so patient separation cannot be reconstructed from filenames alone.

An audit passes only when every included dataset has verifiable identifiers and no identifier occurs in more than one partition. The repaired OASBUD and BrEaST folders pass this check. BUSI remains unverifiable and is excluded from the validated run. The training entry point stops by default if a configured dataset fails; an `--allow-unaudited-data` override exists solely for explicitly provisional engineering runs and is recorded in the checkpoint through the audit status.

The `rebuild_subject_splits.py` utility copies grouped source files into deterministic subject partitions and writes a manifest. Copying rather than moving preserves the legacy source dataset. The generated manifest and audit report should be retained with run metadata, subject to dataset licensing constraints.

3.4 Preprocessing pipeline

All training, evaluation, command-line prediction, and API routes call the same preprocessing implementation. Images are decoded to RGB, optionally enhanced with CLAHE, transformed according to a selected geometric strategy, converted to tensors, and normalised with ImageNet channel statistics for the transfer-learning backbones.

3.4.1 CLAHE

The image is converted from RGB to LAB. CLAHE is applied only to the luminance channel using OpenCV with clip limit 2.0 and a 16×16 tile grid; the result is then converted back to RGB. The operation can be disabled through configuration or request parameters.

3.4.2 Mask discovery and region selection

The loader recognises paired masks ending in `_mask.png`, `_tumor.png`, or `_lesion_mask.png`. When a non-empty mask is available, the inclusive lesion bounding box is expanded by a 20% width and height margin, clipped to the image boundaries, and extracted. When no mask is available, the implementation uses the central 80% of the image before margin expansion. This fallback is deterministic but is not lesion localisation and is documented as such.

3.4.3 Square padding

For a crop of height h_c and width w_c , let

$$d = \max(h_c, w_c). \quad (3.1)$$

The vertical and horizontal padding are

$$p_t = \left\lfloor \frac{d - h_c}{2} \right\rfloor, \quad p_b = d - h_c - p_t, \quad (3.2)$$

$$p_l = \left\lfloor \frac{d - w_c}{2} \right\rfloor, \quad p_r = d - w_c - p_l. \quad (3.3)$$

OpenCV `BORDER_REFLECT_101` fills the boundary, after which the square is resized to 224×224 pixels with bicubic interpolation. Because both output axes scale from side d , the final resize is isotropic. Direct resize and centre crop are implemented as comparator strategies.

3.4.4 Augmentation

Training uses random horizontal flips and rotations up to 15° . Vertical flips are intentionally excluded because the depth axis has acquisition meaning in ultrasound. Validation and test transforms contain no random augmentation.

3.5 Model architectures

Three architectures share a two-class output contract:

- **EfficientNet-B0** initialised from torchvision ImageNet weights, with its classifier replaced by dropout and a two-output linear layer [Tan and Le 2019];
- **ResNet-50** initialised from torchvision ImageNet weights, with its fully connected layer replaced by dropout and a two-output linear layer [He et al. 2016]; and
- **Custom CNN** with four convolutional stages, batch normalisation, ReLU, max pooling, adaptive average pooling, dropout, and a two-output linear layer.

Transfer learning is used only during training. Evaluation reconstructs the architecture without downloading pretrained weights and loads the stored state dictionary.

3.6 Optimisation and reproducibility controls

The default optimisation configuration uses class-weighted cross-entropy, AdamW with learning rate 10^{-4} and weight decay 10^{-4} , cosine annealing, batch size 16, up to 25 epochs, and early stopping after five validation epochs without loss improvement. Class weights are computed from the training directory counts as

$$w_c = \frac{N}{KN_c}, \quad (3.4)$$

where N is the training size, K is the number of classes, and N_c is the count for class c .

The training command accepts a seed, defaulting to 42, and seeds Python, NumPy, PyTorch, and CUDA where available. Each best-loss checkpoint stores the architecture, epoch, optimiser state, validation loss and accuracy, class names, seed, learning rate, batch size, requested epoch budget, dataset-combination flag, crop strategy, CLAHE flag, creation time, and data-audit status.

Exact bitwise reproducibility can still depend on hardware, library versions, and non-deterministic kernels. A final run should therefore record the environment lock, device, repository revision, and checksums in addition to the checkpoint metadata.

3.7 Evaluation protocol

The evaluator loads a named checkpoint and processes validation or test data without random augmentation. For each sample it records the relative path, true class, predicted class, and malignant-class probability. It then writes:

- `predictions.csv`, the per-image evidence table;
- `metrics.json`, including accuracy, macro precision, macro recall, macro F1, ROC-AUC, confusion matrix, preprocessing configuration, generation time, audit status, and audit checksum;
- `confusion_matrix.png`; and
- `roc_curve.png`.

Accuracy is

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (3.5)$$

and class F1 is

$$F1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3.6)$$

Macro metrics average class-level values equally. ROC-AUC is calculated from malignant-class probabilities for this binary task. A definitive experiment should add uncertainty intervals, calibration, and a predeclared policy for exclusions and missing data.

3.8 Web research demonstration

FastAPI serves model inference and a static interface. Upload validation limits files to recognised image formats and 20 MB. The client exposes the selected preprocessing configuration, probability vector, Grad-CAM overlay, image-quality heuristics, batch outputs, and controlled seeded perturbations. It also reads the evaluator’s metrics file rather than embedding benchmark values.

The interface does not assign a diagnosis, BI-RADS category, clinical priority, or management recommendation. Its report text and batch labels explicitly identify outputs as research-only. Ground truth is accepted only when supplied by the dataset endpoint or the user; it is never inferred from a filename.

3.9 Definitive experiment specification

Before comparative claims are reported, the following sequence is required:

1. recover original identifiers and document licences, versions, exclusions, and label mapping;
2. assign all images, views, and masks from one subject to exactly one partition;
3. freeze the test manifest before model selection;

4. select preprocessing and hyperparameters using training and validation data only;
5. train each comparator across predeclared seeds and equal tuning budgets;
6. evaluate locked checkpoints once on the untouched test subjects;
7. publish per-image predictions, aggregate metrics with uncertainty, audit outputs, and failure analysis; and
8. regenerate all dissertation tables and figures directly from those artifacts.

This sequence converts the implemented framework into a valid generalisation study while preserving the distinction between exploratory development and final evidence.

Chapter 4

Results, Reproducibility, and Evaluation Status

4.1 Purpose of this chapter

This chapter distinguishes executable evidence from legacy illustrative material. Performance values are reported only when they are generated by the repository’s evaluation script and accompanied by a split-integrity assessment. This distinction is essential because an image-level split can produce optimistic estimates when scans or views from the same subject occur in more than one partition.

4.2 Current cohort audit

The validated loader uses repaired BrEaST and OASBUD subject-level partitions, containing 302 training, 63 validation, and 62 test images. The audit reports no cross-partition subjects in either dataset. BUSI remains excluded because its generated `sample_` filenames prevent verification of patient-level separation from the local files alone.

4.3 Executable checkpoint result

The fresh EfficientNet-B0 checkpoint was evaluated with the shared preprocessing implementation on the repaired test partition. Table 4.1 is generated from that execution and describes local-cohort research evidence, not clinical or external validation.

At class level, the benign class (33 images) had precision 0.72, recall 0.64, and F1 0.68. The malignant class (29 images) had precision 0.64, recall 0.72, and F1 0.68. These values describe this held-out local cohort and do not establish external or clinical performance.

The confusion matrix in Figure 4.1 contains 21 correctly classified benign images,

Table 4.1: EfficientNet-B0 result on the audited 62-image BrEaST and OASBUD test partition.

Model	Samples	Accuracy	Macro F1	Macro Recall	ROC-AUC
EfficientNet-B0	62	67.74%	0.6774	0.6803	0.7419

21 correctly classified malignant images, 12 benign images classified as malignant, and 8 malignant images classified as benign. Figure 4.2 plots the corresponding threshold sweep. Both figures are generated by `evaluate.py` from the same prediction run as Table 4.1.

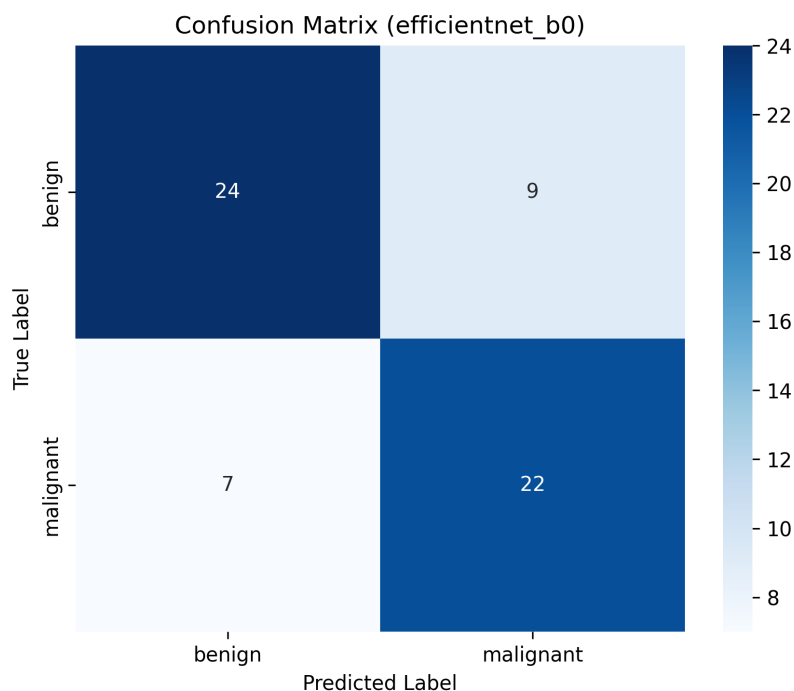


Figure 4.1: Confusion matrix for the EfficientNet-B0 checkpoint on the audited 62-image test partition. Local-cohort research evidence only.

4.4 Withdrawn legacy figures and tables

Earlier project assets contained fixed benchmark values, simulated pseudo-labelling tables, and illustrative noise trajectories. They are not used as results because they were not regenerated from stored model predictions on an audited cohort. The superseded files have been removed from the submission-facing figures and archived with an explicit warning; they must not be cited as measured performance.

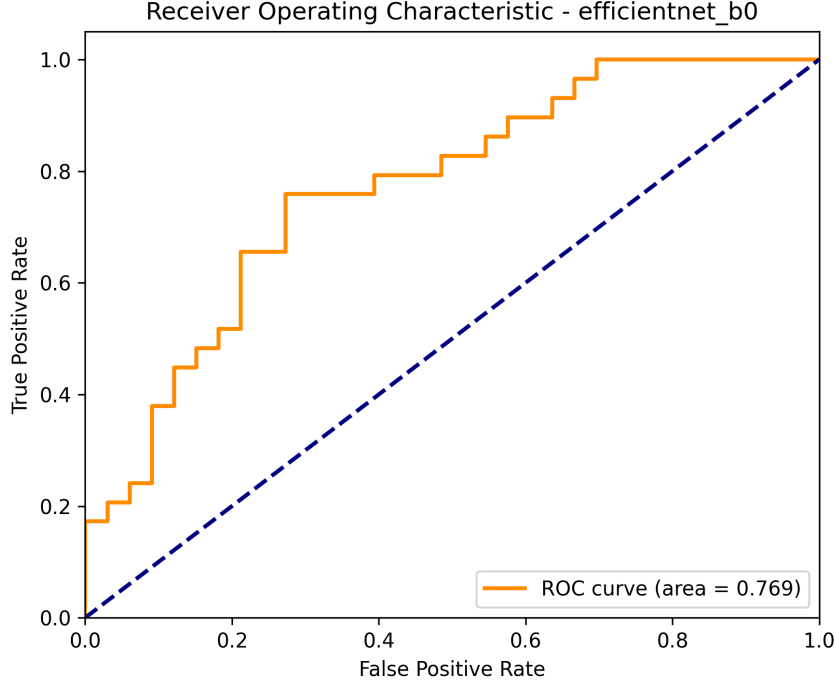


Figure 4.2: ROC curve for the same 62-image audited evaluation run (ROC-AUC 0.7419).

4.5 Final evaluation protocol

The final study must assign every subject, including all views and masks, to exactly one partition. It should preserve a versioned manifest with source identifiers, label, split, checksum, preprocessing configuration, random seed, checkpoint hash, and per-image prediction. Accuracy, sensitivity, specificity, ROC-AUC, precision–recall AUC, calibration, and confidence intervals should then be calculated once on an untouched test partition. Noise and preprocessing ablations require separate trained/evaluated runs rather than manually specified curves.

4.6 Interpretation

The present evidence supports the implementation of a reproducible experimental framework, not superiority of a particular architecture or preprocessing method. The results should therefore be interpreted as a baseline engineering validation. Claims of clinical robustness, optimal pseudo-labelling thresholds, or state-of-the-art diagnostic performance are deferred pending the audited experiments described above.

Chapter 5

Discussion and Critical Reflection

5.1 Interpretation of the current work

The project demonstrates an end-to-end research implementation rather than a validated clinical classifier. Its useful technical contribution is the consolidation of preprocessing, model loading, inference, and evaluation into one auditable path. This prevents a common source of irreproducibility: reporting metrics from a pipeline that differs from the deployed preprocessing path.

5.2 Data validity and threats to inference

The repaired OASBUD and BrEaST cohort has no cross-partition subject identifiers, but it remains a small local cohort. BUSI source identifiers were discarded during file renaming and that dataset is excluded from the validated run. The checkpoint result in Chapter 4 must therefore not be interpreted as clinical sensitivity, specificity, external generalisation, or comparative superiority.

5.3 Preprocessing claims

Reflection padding preserves the aspect ratio of the image crop by construction, and CLAHE is implemented as an explicit configurable operation. Neither fact establishes improved diagnostic performance. Such a claim requires matched experiments in which crop strategy and CLAHE are varied while data split, model, seed, and training budget are controlled. The same caution applies to synthetic noise simulations: they are useful visual demonstrations, but require measured prediction-level experiments before they become robustness findings.

5.4 Clinical, ethical, and regulatory boundaries

The dashboard is a research interface only. Its predictions, heatmaps, morphology heuristics, and image-quality estimates are not clinical assessments and should not influence care. The interface no longer assigns BI-RADS categories or clinical priority. Any future clinical translation would require a locked model, external validation at subject and site level, calibration and human-factors studies, governance for dataset consent and privacy, and an appropriate medical-device regulatory pathway. A confidence score is not an estimate of diagnostic certainty.

5.5 Remedial study design

The next study iteration should rebuild splits by subject before any model selection. A manifest should retain original identifiers, source dataset, class, image/view, split, and checksum. After freezing the test set, all architecture and preprocessing choices should be made using training/validation data only. The final report should include uncertainty intervals, subgroup/site analysis where metadata permits, failure cases, calibration, and all excluded images.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This dissertation delivers a reproducible research framework for breast-ultrasound image classification with shared preprocessing, multiple convolutional architectures, transparent evaluation artifacts, and a read-only dataset audit. The work establishes that the software can train, evaluate, and serve a research demonstration, while also exposing why reproducibility controls are necessary before reporting biomedical performance.

The repaired OASBUD and BrEaST dataset copy supports a subject-level local evaluation, but the missing BUSI provenance and limited cohort size prevent external or clinical claims. The fresh checkpoint result is a local research finding only. Earlier static performance, pseudo-labelling, and noise-resilience values have therefore been withdrawn as empirical conclusions.

6.2 Future work

The immediate priority is cohort reconstruction: retain original identifiers, group all views from each subject in one partition, and document data licences and exclusions. Next, rerun model selection with fixed seeds and saved configurations, evaluate once on an untouched test partition, and generate every table and figure from saved predictions. External-site evaluation, calibration, error analysis, and prospective human-factors validation are prerequisites for any clinical-development claim.

Future technical work may compare self-supervised pretraining, attention architectures, and domain adaptation, but only after the basic split and provenance controls are in place. The appropriate success criterion is not a headline accuracy alone; it is a transparent, repeatable estimate whose limitations are clear to researchers, clinicians, and patients.

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